COMP 3500SEF · Final Report

Warehouse Logistics Management System
Group 42: Tang Yi Jie, Yin Pei Xu, Tang Chao, Yang Ting Zhang, Wu Peng Bo, Chung Ngon Man, Li Mu Qiu.

1. Scope

The goals are traceable order decisions — each order's history is auditable through ledger entries and notifications — and a role-separated workflow. Customers order; staff fulfil; drivers deliver; administrators report. Deployment is local-only: one app (§4); seed data only (RK-10).

2. Development process

An increment is work closed by a review or test run.

Table A. Increments

Increment and date	Trigger	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, responsibilities, contracts recorded (register)
17 Oct review	Stock and order review	Two defects; fixed as row 3
Confirmation, 26–28 Oct 2026	Row-2 defects: repeat submits, no ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	All four workflows integrated
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass (privacy tests)
Diagram review, 10–11 Nov 2026	Diagrams diverged	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three in-app interactions
CI health tests, 11–12 Nov 2026	CI-run outcomes untested	Two regression tests; suite reached 38 (unit.log)

3. Requirements

- FR-01 create orders with lines and totals Test. - FR-02 enforce lifecycle and stock release Test (cancel). - FR-03 number orders and shipments, track owners Test. - FR-04 show inventory, ledger, low-stock alerts Test. - FR-05 record and read order notifications Test. - FR-06 provide admin aggregates and CSV Test. - FR-07 authenticate and enforce access Test. - FR-08 assign orders and record delivery Test.

- NFR-01 performance — API p95 (95th percentile) ≤2 s, first contentful paint ≤3 s (measurement). - NFR-02 CI health — every commit ends in green unit, integration and browser runs (observations). - NFR-03 least privilege — role gates; address only to the driver (privacy tests). - NFR-04 interaction budget — every core operation (§8) takes at most three in-app interactions; one interaction is a fill, selection or click (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run.

Table B. Risk coverage: RK-01–10 (coverage).

4. System models

Four layers separate concerns: pages, routes, handlers, Prisma/SQLite persistence.

Each **order item** references one product; orders hold one or more items and one user; an order has at most one shipment recording events; products accumulate movements (InventoryLog); notifications relate to at most one order.

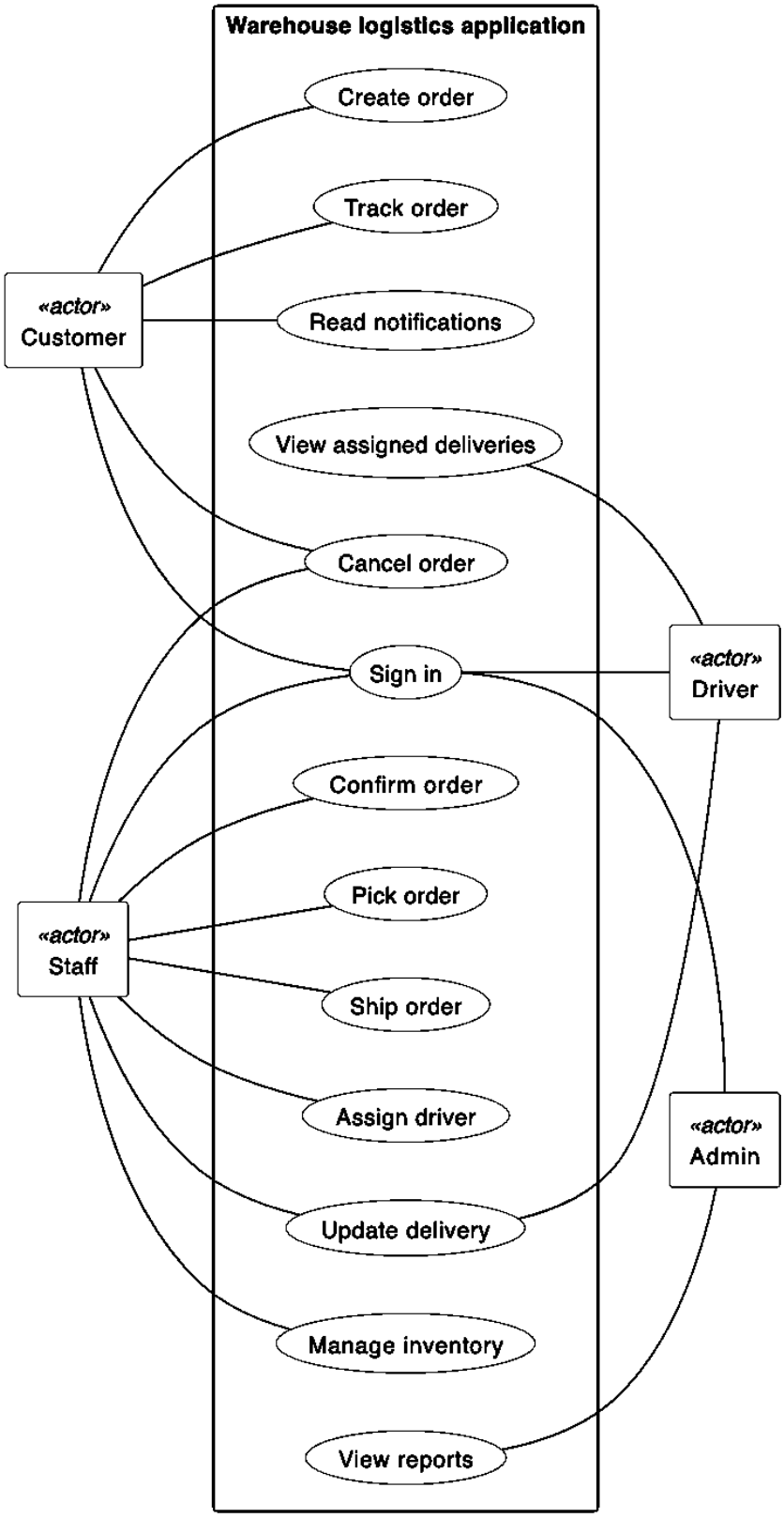


Figure 1. Role functions: four actors, thirteen use cases.

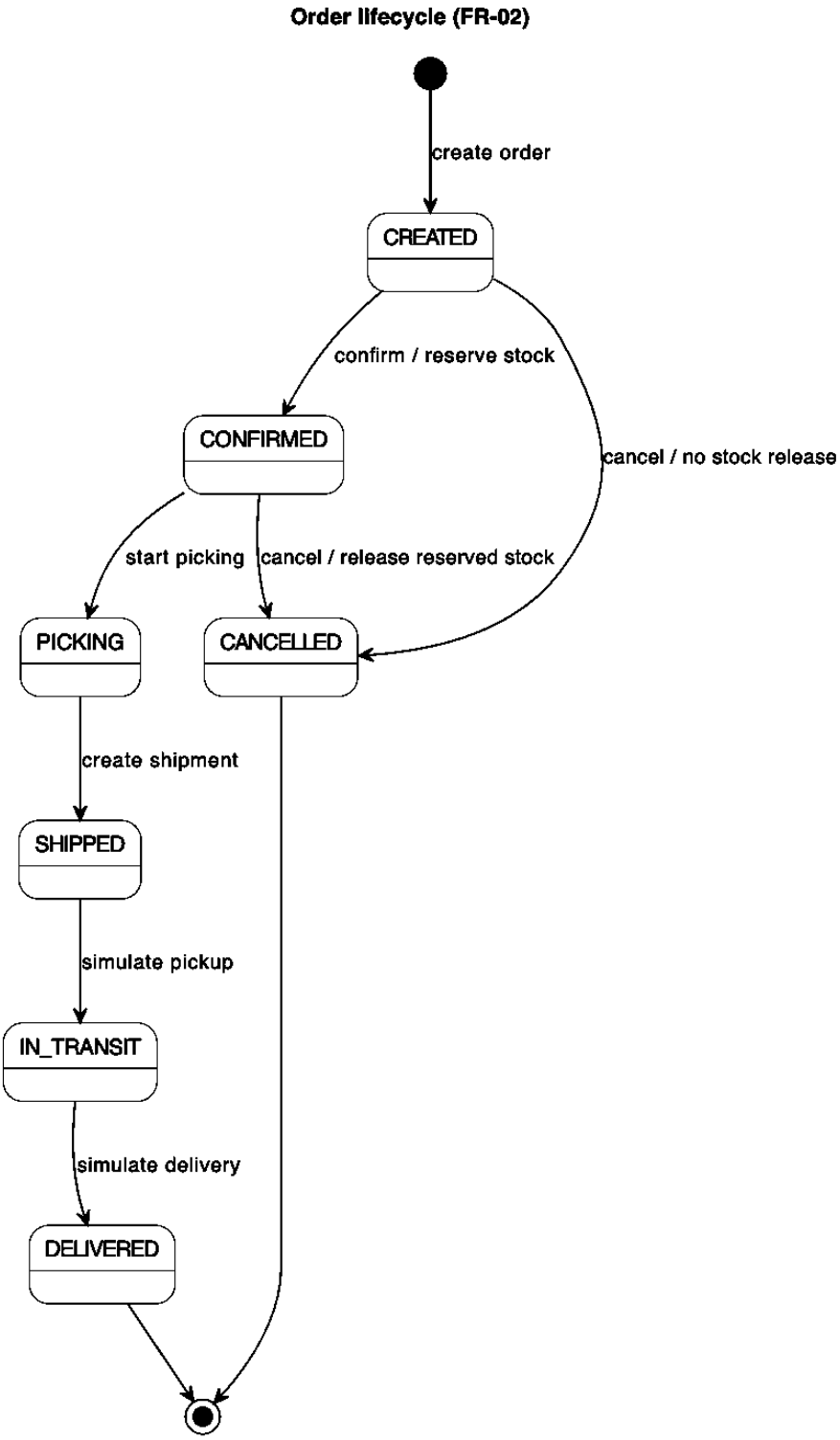


Figure 2. Order transitions: reserve, release or neither.

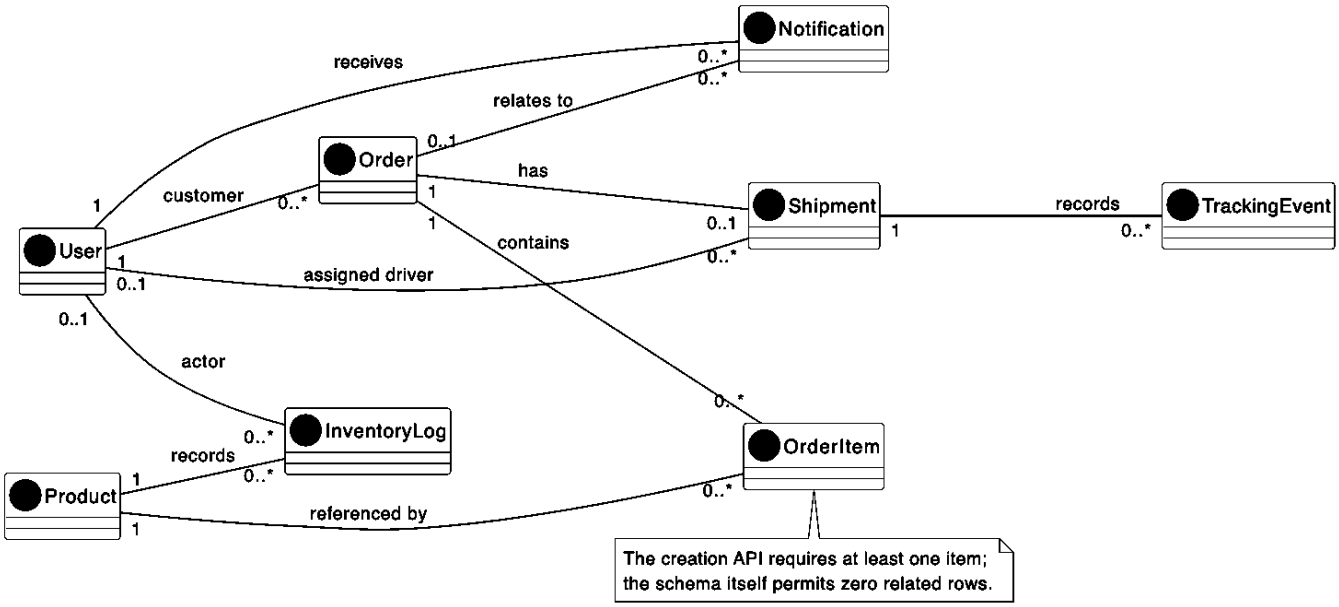


Figure 3. Data relationships: entities and multiplicities as an ER model.

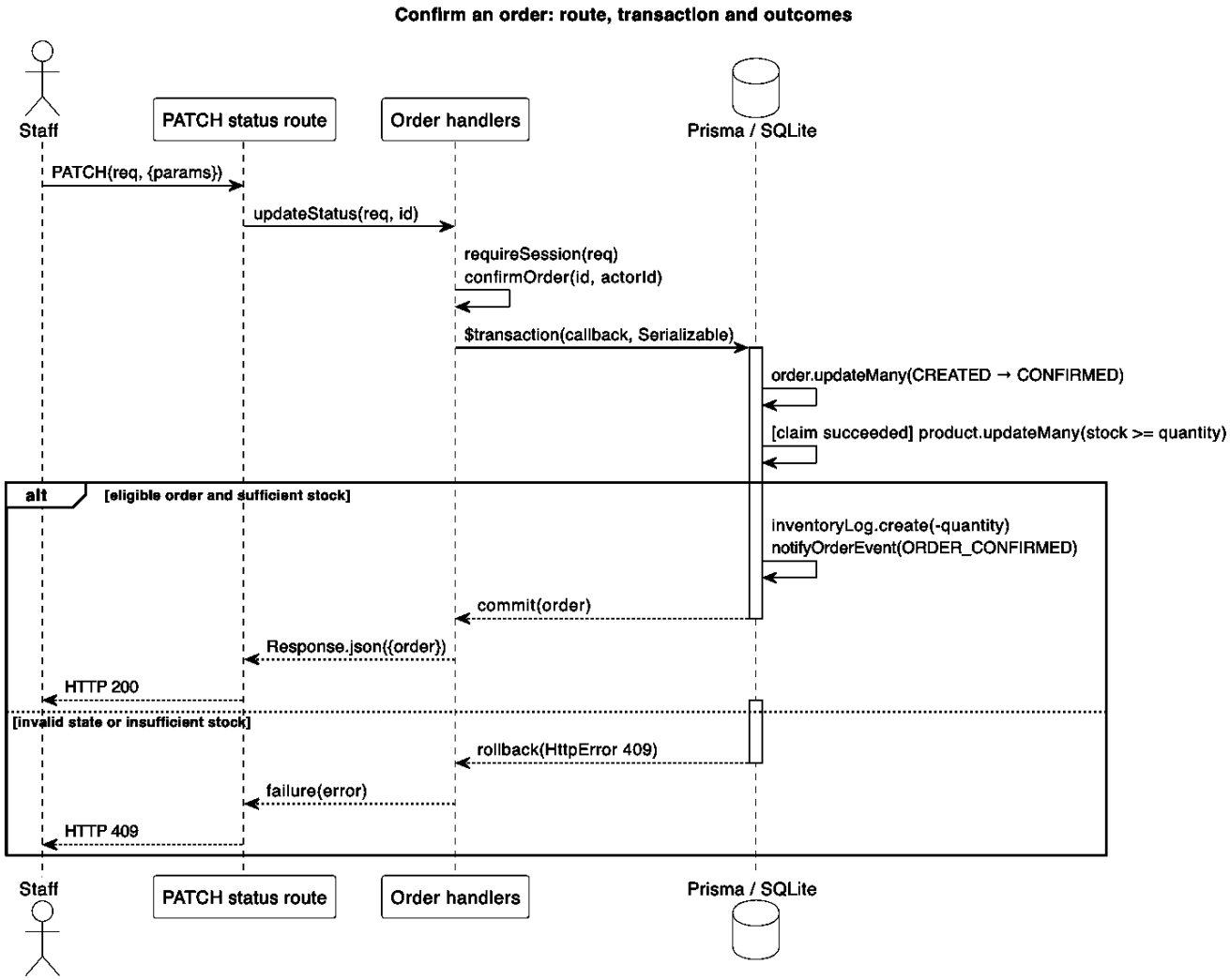


Figure 4. Confirmation transaction: guard, conditional writes, rollback on failure or short stock.

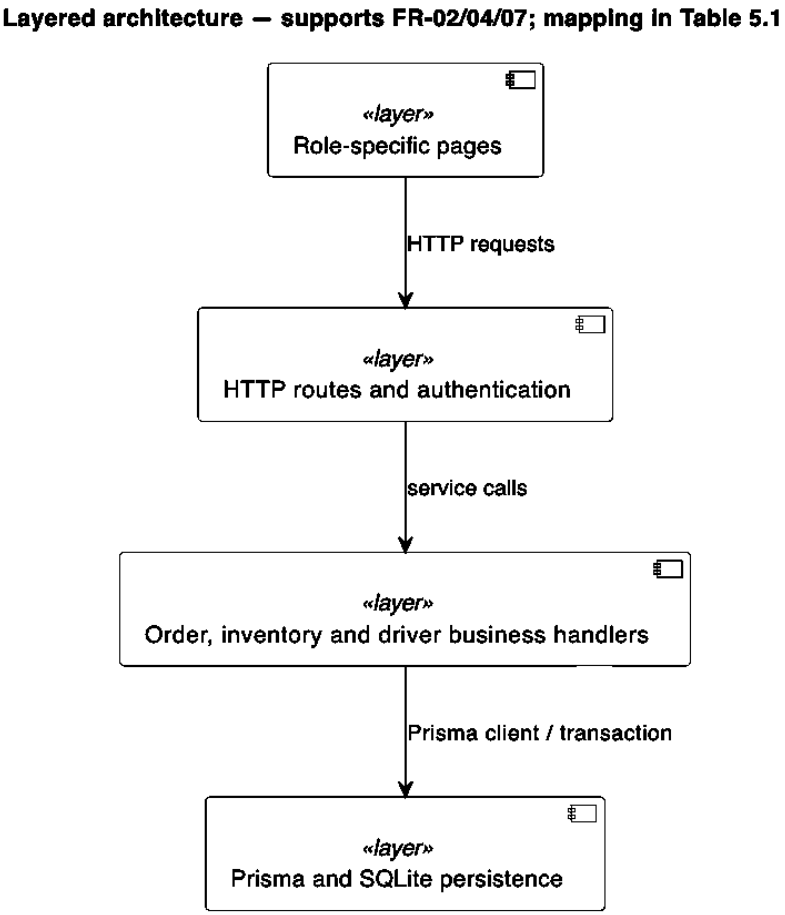


Figure 5. Four layers: pages, routes, handlers, Prisma/SQLite.

5. Implementation

Confirmation reserves stock. Lifecycle: CREATED → CONFIRMED → PICKING → SHIPPED → IN_TRANSIT → DELIVERED; CANCELLED from CREATED or CONFIRMED.

Role	Transition and effect
Customer	CREATED — cancel, no restock
Customer	CONFIRMED — cancel, restock
Staff	CREATED — rejected
Staff	CONFIRMED — cancel, restock
Customer, staff	PICKING+ — rejected

A CREATED order stays with the customer; staff have no refuse action (§1); the lifecycle is locked as test constants (tests).

Design	Implementation and verification
Confirmation: atomic; rollback on short stock; repeats rejected (HTTP 409)	order-api.ts:245; Orders
driverId: SHIPPED only, kept after delivery	driver-api.ts:44; Driver
FR-03 numbering and tracking; FR-04 ledger, alerts, replenishment	create-order.ts:91, inventory-api.ts:142; Inventory

6. Verification

CI records 38 unit, 140 integration checks; 2 integrity checks guard the evidence chain: the demo-data reset needs RESET_DEMO_DATA=YES and the bundle's SHA-256 manifest must match the source (integrity checks). Logs: integration "tests 140", "pass 140"; unit "Tests 38 passed (38)"; browser "Browser acceptance passed: 26 checks".

Loopback measurement (11 Nov): API p95 3.2–13.6 ms, first contentful paint 64–96 ms against 2 s/3 s; not representative of public networks.

Reviews (Table A rows 2 and 6) verified requirements and models; row 7's fix checked via the interface (§8).

7. Evolution

Three reviews drove changes: 17 Oct found two defects — repeat submits and a missing ledger — fixed by §5; 10–11 Nov aligned diagrams, traceability and rewrote NFR-01/02, the window (08:00–24:00) becoming CI-run checks; 11 Nov added the direct export path (issue #15); two tests followed as row 8 (36→38).

8. Validation

The automated walkthrough covers eleven core operations (of the thirteen use cases in Figure 1) — create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates — at most three in-app interactions each; pickup and delivery are recorded in-app. "Browser acceptance passed: 26 checks" — 20 operation checks (5 once, 3 twice, 3 thrice) plus 6 cross-cutting (two each: input validation, session handling, interface). No application code changed after this run; row 8 added tests.

9. Feedback and reconciliation

Table C. Feedback to evidence

Feedback (source, date)	Disposition and evidence
Diagrams diverged (10–11 Nov, corrections)	Corrected
Stock effects under FR-04 (17 Oct, traceability)	Moved to FR-02
Thin register triggers (10–11 Nov, register)	Rewritten
Filter-then-export took four interactions (11 Nov, issue #15)	Combined path rejected; date selection meets budget; issue closed
No CI-health test (11 Nov, log)	Suite reached 38
NFR-02 daily availability window (10–11 Nov, revision log)	CI-run checks (§3)

10. Evidence

38 unit / 140 integration / 2 integrity / 26 browser checks (11 Nov) (BASELINE, traceability).